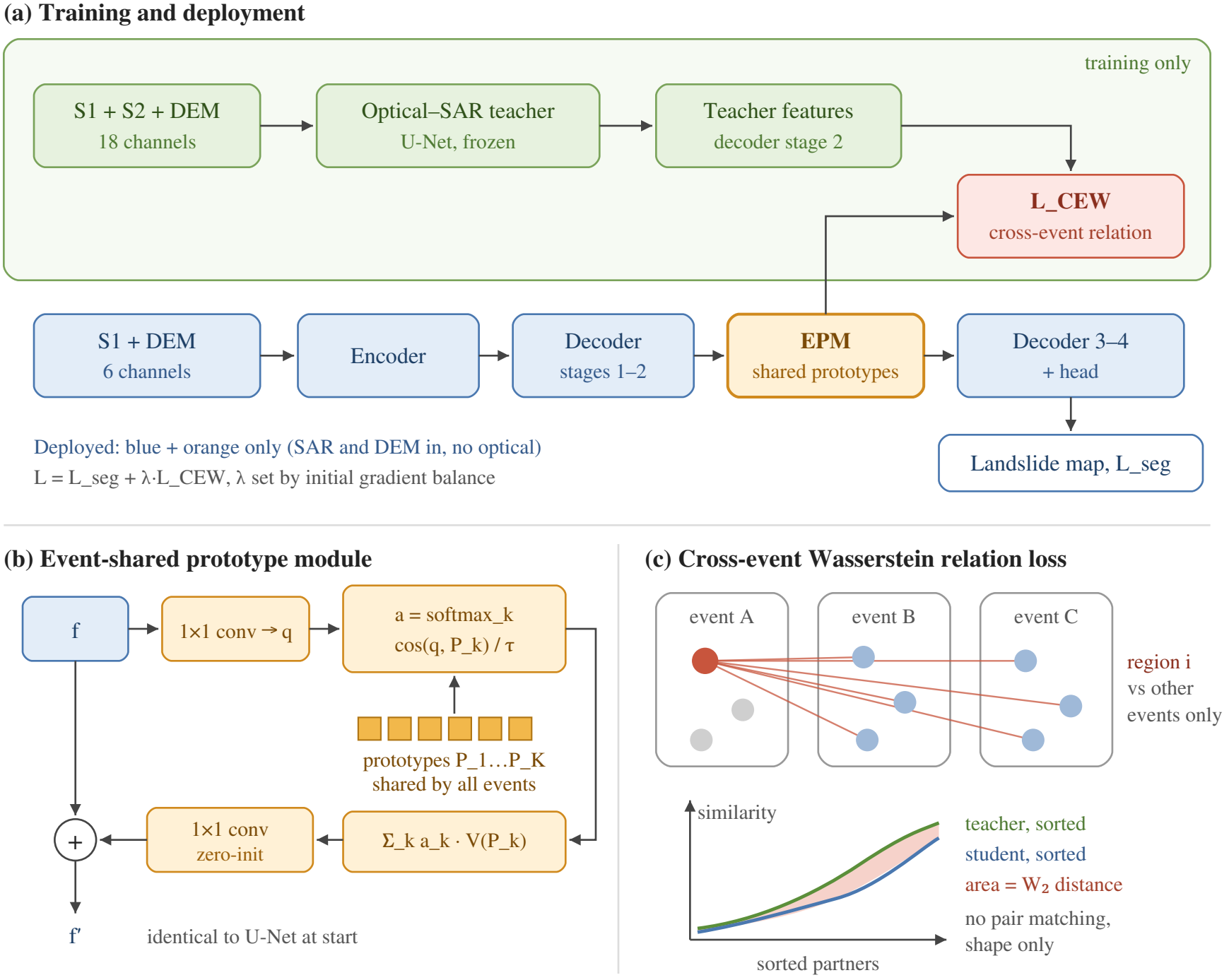

(a) Training and deployment
training only
S1 + S2 + DEM
18 channels
Optical–SAR teacher
U-Net, frozen
Teacher features
decoder stage 2
L_CEW
cross-event relation
S1 + DEM
6 channels
Encoder
Decoder
stages 1–2
EPM
shared prototypes
Decoder 3–4
+ head
Landslide map, L_seg
Deployed: blue + orange only (SAR and DEM in, no optical)
L = L_seg + λ·L_CEW, λ set by initial gradient balance
(b) Event-shared prototype module
f
1×1 conv → q
a = softmax_k
cos(q, P_k) / τ
prototypes P_1…P_K
shared by all events
1×1 conv
zero-init
Σ_k a_k · V(P_k)
+
f′
identical to U-Net at start
(c) Cross-event Wasserstein relation loss
event A
event B
event C
region i
vs other
events only
similarity
teacher, sorted
student, sorted
area = W₂ distance
no pair matching,
shape only
sorted partners